

An unusual osteoma in the mandibular condyle and the successful replacement of the temporomandibular joint with a custom-made prosthesis: a case report.

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An osteoma is a benign tumor of bone with unknown etiology and is considered rare, mostly restricted to the craniofacial skeleton. This case report describes an uncommon condylar osteoma in a 67 years old white female patient with laterognathism to the left side, limited mouth opening, aesthetic change and pain associated with the right temporomandibular joint (TMJ). The histopathological examination confirmed osteoma. The lesion was surgically excised and immediate reconstruction was carried out using a custom-made total TMJ prosthesis. The patient has been in follow-up for 2 years, with no symptoms. Unilateral total TMJ prosthesis can be considered to replacement of TMJ after osteoma excision with resection of the condyle.

Analysis of outcomes after revision replacement of failed total temporomandibular joint prostheses.

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End-stage disease of the temporomandibular joint (TMJ) can be managed successfully with alloplastic total replacements, but these can fail because of allergy, infection, wear, fracture, and heterotopic development of bone. We prospectively reviewed the outcome data of all patients who required revision of previously placed replacement joints between 2004 and 2016. Data included pain and diet scores using 100mm visual analogue scales (VAS), and interincisal distance recorded before, and at six weeks, six months and 12 months after operation. The reasons for failure and the number of previous procedures were also noted. Twenty patients (26 joints) had revisions. The reasons included infection (n=9), reankylosis (n=5), wear of the existing prosthesis (n=2), fracture of the prosthesis (n=2), foreign body reaction (n=1), and allergy to the prosthesis (n=1). The mean (range) age of the patients was 53.3 (47-68) years, and 15 were female and five male. Preoperatively, the mean (SD) pain score was 73.1 (22.4), mouth opening was 20.9 (10.2) mm, and diet score 41.7 (23.6). At 12-month follow up, all the measurements had improved significantly ($p \leq 0.05$), with the pain score improving to a mean (SD) of 18.4 (25.2), mouth opening to 32.2 (9.3) mm, and diet score to 89.4 (18.5). Revision replacements done by an experienced team result in considerably improved outcomes with limited complications, but the improvements in function and pain are not as marked as they are after primary replacement.

Alloplastic Temporomandibular Joint Replacement in Patients With Systemic Inflammatory Arthritis and Connective Tissue Disorders.

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We present a retrospective study to report the outcomes of total temporomandibular joint (TMJ TJR) replacement with alloplastic devices in patients suffering from systemic inflammatory arthropathies. A total of 39 patients with a diagnosis of rheumatoid arthritis (RA), ankylosing spondylitis (AS), psoriatic arthritis (PA), juvenile idiopathic rheumatoid arthritis (JIA), or systemic lupus, underwent alloplastic total joint replacement(s) (TJR) from 1999 to 2019. Maximal interincisal opening (in mm) was recorded before surgery (T0), at 1-year post-surgery (T1) and at last follow-up (T2). A visual analog scale (from 0 to 10)

was used for subjective examination of pain before surgery (T0) and at last follow-up (T2). Comparisons were conducted with statistical significance set at $P < .05$. Seventy-four joints were replaced in 39 patients. Thirty-two were female. The mean age was 36 years old (range 18-61) and the mean follow-up was 45.9 months (SD 49.4). The most common diagnosis was RA ($n = 21$), followed by JIA ($n = 5$) and AS ($n = 5$), PA ($n = 4$), lupus ($n = 3$), and mixed connective tissue disorder ($n = 1$). The mean pain score had fallen from 6.8 (SD 3.2) at T0 to 1.3 (SD 2.4) ($P < .001$) at T2. The maximal interincisal opening had improved from a mean of 22.1 mm (SD 13.3) at T0 to 34.3 mm (SD 8.5) ($P < .001$) at T2. One patient got persistent dysesthesia in the V3 distribution. There were no serious late complications. Patients suffering from systemic inflammatory arthropathies involving the TMJs can be successfully treated by TJR with alloplastic devices. The long-term reduction of TMJ symptoms and functional improvement in this initial study suggest good predictability for this treatment.

Total Joint Replacement After Condylar Destruction Secondary to Long-Standing Dislocation of the Temporomandibular Joint.

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To describe the treatment of 2 long-standing chronic dislocation of the temporomandibular joints (TMJs) and 1 chronic recurrent dislocation. This report describes the treatment of 3 patients; 2 with a long-standing chronic dislocation of the TMJ and 1 with a chronic recurrent dislocation. Duration of dislocation and anatomical considerations make the treatment challenging and controversial. The patients presented in this report all developed destruction of their condyles. They were successfully treated with total joint replacement with alloplastic devices. All the 3 patients underwent successful surgery and recovery. Mandibular function and pain level were significantly improved. Long-standing dislocation of the TMJ is rare. This condition can be successfully treated by resection of the damaged condyles and reconstruction with alloplastic total TMJ replacements.

Total temporomandibular joint replacement: a clinical case with a proposal for post-surgical rehabilitation.

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The literature on total alloplastic temporomandibular joint (TMJ) reconstructions is encouraging, and studies on total alloplastic TMJ replacements' outcomes showed acceptable improvements in terms of both pain levels and jaw function. Nevertheless, a better standardization of both surgical and post-surgical phases should further improve the efficacy of treatment. The case report describes the surgical and post-surgical phases of treatment in a patient who underwent a total unilateral joint replacement for TMJ ankylosis, along with a proposal for a post-surgical rehabilitation protocol providing strong passive and active physiotherapy and hyaluronic acid injections to the contralateral joint. The post-operative (PO) course was uneventful. The patient showed a marked improvement up to about 60% in mouth opening and had no pain on either side at the 1-year follow-up. In a patient with unilateral TMJ ankylosis, total TMJ replacement was successful. A carefully tailored post-surgical rehabilitation protocol helped the patient to gain a clinically significant improvement in jaw function. Longer follow-up periods are needed to assess the long-term maintenance of results. Clinical trials are strongly recommended to assess the relative efficacy of different PO protocols.

The compressible silicone rubber prosthesis in temporomandibular joint disease.

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An alternative technique for temporomandibular joint arthroplasty is described, in which the mandibular condyle is replaced by a soft compressible silicone rubber prosthesis. A modified Nicolle-Calnan metacarpo-phalangeal joint prosthesis was used to reconstruct 31 joints in 24 patients. Results suggest that in those patients where there has been no loss of the prosthesis, function continues to be markedly improved when compared with the pre-operative condition. Painful symptoms were relieved in cases of specific joint pathology but the technique was of little value in the management of dysfunctional pain when radiographic evidence of joint pathology was absent. A specifically designed TMJ prosthesis of this type may be a useful addition to the surgeon's armamentarium.

Vascular metatarsophalangeal to ankylosed temporomandibular joint replacement.

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Four cases of unilateral temporomandibular (TM) joint ankylosis were successfully treated by metatarsophalangeal to TM joint replacement by microvascular technique. Results 6 to 20 months postoperatively were encouraging. Further follow-up studies are necessary to evaluate the long-term results.

Inverted, T-shaped silicone implant for the treatment of temporomandibular joint ankylosis.

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Reconstruction of the ankylosed temporomandibular joint is a challenging task. Speech impairment, difficulties with mastication, poor oral hygiene, facial asymmetry, and mandibular micrognathia results in physical and psychologic disabilities. Various surgical techniques with varying success rates have been reported. Many autogenous and alloplastic materials have been proposed. The authors used an inverted, T-shaped silicone implant for the reconstruction of the temporomandibular joint after the release of the ankylosis in 10 patients without any complications in the postoperative period. The authors assert that the reconstruction of the ankylosed temporomandibular joint with an inverted, T-shaped silicone implant is a reliable and effective alternative. This technique can be used according to the special requirements of each patient and obviating the need for the fixation of the implant and is a safer and better way of using silicone for the treatment of temporomandibular joint ankylosis.

The goat as a model for temporomandibular joint disc replacement: Techniques for scaffold fixation.

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A state-of-the-art scaffold capable of efficiently reconstructing the temporomandibular joint (TMJ) disc after discectomy remains elusive. The major challenge has been to identify a degradable scaffold that remodels into TMJ disc-like tissue, and prevents increased joint pathology, among other significant complications. Tissue engineering research provides a foundation for promising approaches towards the creation of successful implants/scaffolds that aim to restore the disc. In light of improving the quality of life of patients who undergo TMJ disc removal, it is critical to establish a preclinical animal model to evaluate the properties of promising scaffolds implanted post-discectomy and to determine the most efficient implantation procedures to ensure a more reliable in-depth evaluation of the biomaterial replacing the articular disc. The present study evaluated the outcomes of two protocols for implantation of an acellular scaffold composed of an extracellular matrix (ECM) derived from the small intestinal submucosa (SIS) of the pig, as a regenerative template for the TMJ disc in a goat model. The outcomes suggest that leaving one-half of the disc medially will allow anchoring of the device to the medial aspect of the joint while avoiding lateral displacement of the ECM scaffold. The goat model is ideal to assess the longevity of tissue-engineered solutions for the TMJ disc, considering that goats chew for 12-16 hours a day. This study provides an important reference for the selection of a suitable scaffold implantation procedure and the goat model for the development of new strategies to assess TMJ disc regeneration.

TMJ Concepts Patient-Fitted Temporomandibular Joint Reconstruction Prosthesis System: Results From a Food and Drug Administration Postmarket Surveillance Prospective Cohort Study.

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Custom-designed and patient-fitted temporomandibular joint replacement (TMJR) devices are used to manage end-stage temporomandibular joint (TMJ) pathology, distorted bony TMJ architecture, or in patients who have undergone multiple prior failed TMJ surgeries. This study aimed to measure the frequency of revision or replacement over time, estimate the 1-, 2-, and 3-year survival rate of implants, and determine what demographic, anatomical, or operative variables may be associated with a device adverse event (AE). A prospective cohort study was conducted on consenting adults who could read English, implanted with the TMJ Concepts TMJR prostheses between 2013 and 2015 by private or academic surgeons. Patients under the age of 18 and those who were incapable, unwilling, or unable to complete the perioperative forms were excluded. The primary outcome variable was the frequency of AEs, which is defined as device revision or replacement. The secondary outcome variable was the survival rate for the TMJ Concepts devices. Device survival at 1, 2, and 3 years was estimated using the Kaplan-Meier methodology. Cox proportional hazards regression was used to analyze the effects of the predictor variables for the device survival rate. A P value of $<.05$ was considered significant. A total of 738 subjects (1,098 joints) were enrolled in the study with 102 (14%) males and 636 (86%) females. The mean age was 45.4 (+14.8) years. The median follow-up period was 36.2 months (95% CI: 36.1 to 36.2 months). A total of 187 subjects (25.3%) were lost to follow-up during the study period. Forty-nine devices (4.46%) were associated with an AE at 3 years. The 3-year device survival was 95.1% (95% CI: 93.6 to 96.3%). The only significant risk factor for an AE was the asymmetry of the mandible (hazard rate ratio = 1.989, $P = .03$). The primary diagnosis was not associated with the time to an AE (P value = .8685). Infection (44.7%, 21 joints) and material sensitivity (12.8%, 6 joints) were the most common reasons specifically for device removal/replacement in the study cohort. Infection was the primary reason reported for device revision or replacement followed by material sensitivity. In the study cohort, the device survival following implantation of the TMJ Concepts prosthesis was not significantly different than reported by the TMJR stock device.

